Project Synopsis

Student Introduction

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Course: B.Sc. Computer Science (Third Year)

This project, titled "Multiple Disease Prediction Using Machine Learning and Streamlit," has been developed as part of the academic curriculum to demonstrate practical application of machine learning techniques in healthcare. The goal is to build an accessible and reliable web tool to assist in early diagnosis of chronic diseases.

Project Title:

Multiple Disease Prediction Using Machine Learning and Streamlit

Project Goals and Objectives:

The primary objective is to develop a machine learning-based web application capable of predicting multiple diseases-Heart Disease, Diabetes, and Parkinson's Disease-by analyzing user-input health data.

Key Objectives:

- Enable early diagnosis and timely medical intervention.
- Provide a user-friendly, accessible interface using Streamlit.
- Offer personalized recommendations related to diet, medications, and lifestyle changes.

Problem Statement:

Millions of individuals suffer from undiagnosed chronic diseases due to late detection, leading to complications and increased healthcare burdens. There is a critical need for:

- A tool that analyzes health metrics accurately.

- A platform capable of predicting multiple diseases in one place.

- An interface that is simple enough for non-technical users to operate.

This project aims to bridge that gap with a smart, accessible, and efficient prediction system that supports both patients and medical professionals.

Tools and Technology Stack:

Languages and Libraries:

- Python: Core programming language.

- Pandas, NumPy: Data preprocessing and manipulation.

- scikit-learn: ML algorithm development.

- Matplotlib, Seaborn: Data visualization.

Machine Learning Algorithms Implemented:

- Support Vector Machine (SVM)

- Random Forest

- Decision Tree

- K-Nearest Neighbors (KNN)

- Logistic Regression

Other Tools:

- Kaggle Datasets: For training and validation.

- Pickle: Model serialization and saving.

- Streamlit: Frontend deployment.

- Streamlit Cloud: Hosting and deployment.

Expected Outcome:

The final deliverable is an interactive web application that:

- Allows users to select a disease category.

- Collects relevant health parameters.

- Predicts whether the user is at risk for the selected disease.

- Provides actionable health guidance based on prediction results.

Additional Benefits:

- High prediction accuracy (e.g., up to 94% for Parkinson's).

- Real-time feedback on user input.

- Enhanced accessibility for early diagnosis and self-monitoring.